

4K Resolution Revolutionising The Television World

4K, also known as ultra high-definition technology, is advanced but there is still some time before products using this technology find acceptance in the global market

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The era of high-definition content is over. It is time for ultra high-definition (UHD) content to take over. Television segment is rapidly becoming a hot-spot of technology advancements. The evolution of 4K UHD TV is an example. Over the last couple of years, the demand for high-definition (HD) content has grown double-fold, aided by the growth in demand for HD LCD and LED TVs.

The most important factor that drives the television industry and generates customer interest is the display. Ever since flat-panel television hit the mass market, the product has been successful in striking the right chord with consumers asking for large display sizes.

The latest in the line of technology offerings ready to hit the television industry is the 4K or Ultra High-Definition (UHD) segment. UHD TV displays are said to bring four times the resolution of 1080p, which is why these are termed as 4K. 4K UHD is an advanced display resolution technology on display panels capable of carrying and presenting native videos at a resolution of 3840×2160 pixels. This is four times the full HD resolution of 1920×1080p, and is achieved by doubling each pixel horizontally and vertically on the full HD display. The result is visibly sharper, crisper and more detailed images than ever before.

It's just the beginning!

4K televisions are available in the Indian market, but not many players have ventured in this segment so far.



LG's 84LM9600 4K TV

LG and Sony are pioneers in terms of launching their 4K TVs in the Indian market.

4K TV is still a niche concept, which will take time to gain acceptance amongst masses. Currently, there is almost nil 4K content available. Also, channels are reluctant to broadcast UHD images due to the significant investment necessary in upgrading the entire image capturing and broadcasting systems.

But one can be hopeful that things will change sooner than later. TV makers think that 4K will significantly increase the perceived image quality for consumers on large screens. This will enable a more immersive experience by facilitating closer viewing distances. However, this format is just being developed for Digital Cinema applications, and further developments and standards for broadcast production environ-

ments are required. In distribution, 4K will require a new compression format called high-efficiency video codec (HEVC), which will hit the market in three to five years.

According to a report titled "What Follows HDTV? A Status Report on 1080p/50 and 4K," consumer flat-panel displays with 4K resolutions will percolate into the market as high-end models and they may potentially be well served with a 1080p/50 distribution format. Alternatives to linear broadcasting, such as downloads, HbbTV (hybrid broadcast broadband TV) and IPTV (Internet Protocol TV), will allow native 4K delivery to the home.

4K TVs: An expensive proposition!

According to a Deloitte report, about twenty 4K TV set models from more than ten vendors should be available by year-end globally. So what is the